The Boundaries of Evidence-Grounded Trust in Multi-Agent Scientific Debate, with a Controlled Benchmark for Sycophantic Consensus

Md. Atikur Rahaman[a,*], Rakibul Hasan[a], Md. Salman Rohoman Nayeem[a], Pratay Paul[a], Yousuf Kamal Himel[a], Mst. Farjana Akter Limu[a], Mohammad Nurul Huda[a] and Ohidujjaman[a]

[a]*Computer Science and Engineering, United International University, Dhaka 1212, Bangladesh*

ARTICLE INFO

Keywords:
Large language models
Multi-agent debate
Retrieval-augmented generation
Sycophancy
Scientific question answering
Trust calibration

Abstract

Large language models can answer scientific questions fluently, but a multi-agent debate over those answers can end in sycophantic collapse: a correct minority agent abandons its answer under pressure from a confident but incorrect majority. Recent systems already check claims against retrieved evidence and reweight agent influence, so the open question is not whether evidence or trust can be used; it is when claim-level evidence-derived trust changes the collective outcome, and where the verifier-accuracy and evidence-redundancy limits lie. This paper states four boundary conditions and specifies the design of a controlled benchmark for the boundary. We describe a deliberation framework that decomposes each agent answer into atomic claims, retrieves scientific passages for each claim, assigns supported, contradicted, or unverifiable verdicts through a verifier interface, and maintains a bounded trust weight per agent; the weight is accumulated in the log domain, mapped through one softmax, and projected onto a simplex with a floor and a ceiling; support and harm are normalized by the number of claims checked; the final answer is a trust-weighted plurality, which reduces to majority voting when the weights are equal. The planned evaluation is a pre-registered 24-run Core matrix on GPQA-Main and MMLU-Pro (primary), with HLE as a ceiling control, and a 32-row Full target with the gold-evidence oracle on EG-SSS (36 rows with the gold-answer oracle as an alternate), with placebo and oracle controls. External evidence is independent of self-reported confidence; B8-no-trust removes the trust signal and the trust-weighted aggregation together, B4-matched isolates the evidence channel, and the boundary analysis states four conditions: equal weights recover majority voting; a verification-accuracy threshold $q^\star$ separates correction from amplification; a redundancy boundary $\rho^\star$ limits what more verification can recover; and a shared-error floor bounds the selective risk that any independent verifier can certify. The current contribution is the framework, the boundary analysis, and the pre-registered protocol: the 13-check gate passes all 13 checks, the current suite passes 615 tests (514 at the P0 release), and Gate 0 completed 10 of 10 vanilla debates with 90 of 90 non-empty positions. The confirmatory matrix is specified but not yet run, so this version makes no claim that the mechanism reduces sycophantic collapse; it reports engineering evidence, not an efficacy result.

1. Introduction

Large language models can answer scientific questions fluently, but fluency is not correctness. Multi-agent debate improves factuality over a single pass by letting several agents read and revise one another's reasoning [1], and chain-of-thought prompting exposes the intermediate steps that the debate evaluates [2]. The final decision is usually a majority vote in which every agent has equal weight [3]. That rule treats repetition as evidence: a confident wrong majority can pressure a correct minority agent into abandoning its answer, and the resulting agreement is recorded as consensus [4]. We call this sycophantic consensus, or sycophantic collapse, and it is the failure this paper studies.

Existing work attacks parts of this failure without changing the aggregation rule. Prompt-level methods such as ConsensAgent detect copying and stalling, then rewrite the task prompt, but the final decision still relies on self-reported confidence [4]. Confidence and uncertainty methods such as DebUnc weight messages from token-level uncertainty, and ReConcile weights models by internal confidence, but neither checks the truth of a claim against

*Corresponding author: Md. Atikur Rahaman (mrahaman2310298@bscse.uiu.ac.bd)

mrahaman2310298@bscse.uiu.ac.bd (Md.A. Rahaman); rhasan2310530@bscse.uiu.ac.bd (R. Hasan); salmanrahman339@gmail.com (Md.S.R. Nayeem); ppaul2310163@bscse.uiu.ac.bd (P. Paul); yhimel2310526@bscse.uiu.ac.bd (Y.K. Himel); mlimu2310535@bscse.uiu.ac.bd (Mst.F.A. Limu)

external evidence [5, 6]. Debate-trigger methods such as iMAD decide whether a debate is needed, but keep the majority vote once debate starts [7]. Static aggregation methods such as Mixture-of-Agents combine model outputs without per-agent evidence or trust [8]. Post-hoc methods such as Minority Sentinel detect a suppressed correct minority after the debate and flip the majority, which is too late to protect the minority during the rounds [9]. Across these studies, influence is set by a vote, a self-reported score, or an internal uncertainty signal; none ties an agent's influence to claim-level external evidence inside the debate. The open question is therefore not whether evidence or trust can be used, but when claim-level evidence-derived trust changes the collective outcome, and where verifier accuracy and evidence redundancy set the boundary.

This paper asks when evidence-grounded trust corrects a wrong majority rather than amplifying it, and how verifier accuracy and evidence redundancy place that boundary. We propose a deliberation framework that decomposes each agent answer into atomic claims, retrieves scientific passages for each claim, assigns supported, contradicted, or unverifiable verdicts through a verifier interface, and maintains a bounded trust weight per agent. The trust weight is accumulated in the log domain, mapped through one softmax, and projected onto a simplex with a floor and a ceiling; support and harm are normalized by the number of claims checked. The final answer is a trust-weighted plurality, which reduces to majority voting when the weights are equal. The planned evaluation is a pre-registered 24-run Core matrix on GPQA-Main and MMLU-Pro (primary), with HLE as a ceiling control, and a 32-row Full target with the gold-evidence oracle on EG-SSS (36 rows with the gold-answer oracle as an alternate), with placebo and oracle controls.

The scope follows from the design. The framework targets scientific question answering, where claims can be checked against a literature corpus and objective answers allow final accuracy and correct-minority retention to be measured. Three heterogeneous model families run as served inference behind an OpenAI-compatible interface, and no model is fine-tuned, so the method is designed to transfer to any model that can be served under the same interface. GPQA-Main [10] and MMLU-Pro [11] form the primary comparison; HLE is a ceiling control; and EG-SSS is a planned gold-evidence set for the claim-level test. The study stops at question types where an answer can be verified and an external corpus exists; open-ended and opinion domains remain outside the scope. The contribution is a bounded, evidence-grounded trust mechanism, four boundary conditions, and a pre-registered evaluation and injection protocol (validation pending) that the community can reuse to measure sycophantic collapse. The current evidence is engineering only: the 13-check gate passes 13/13, the suite passes 615 tests (514 at the P0 release), and Gate 0 completed 10/10 vanilla debates with 90/90 non-empty positions. The confirmatory matrix is specified but not yet run, so this paper makes no efficacy claim.

1.1. Related Work and Positioning

Surveys map the design space of multi-agent LLM systems and its open challenges [12]. Three lines of work are closest to this study. The first is in-debate influence reweighting. DebUnc reweights messages from token-level uncertainty [5]. R^2-MAD estimates per-agent reliability from retrieved debate experience and uses confidence weights to modulate peer influence under a shared misconception [13]. To Trust or Not to Trust maintains message-level and agent-level numeric trust under malicious inputs [14]. Zhu et al. show that diversity-aware initialization and confidence-modulated updates move debate toward truth [15]. Kasprova et al. reduce sycophancy-prone peers' influence with peer sycophancy rankings [16]. Ebrahimi et al. learn credibility scores from past contributions and use them to weight aggregated outputs, remaining effective when adversaries are the majority [17]. These systems establish that in-debate reweighting is feasible. They do not ground the weight in claim-level external scientific evidence, and confidence-based reweighting inherits the calibration error of the underlying model [18].

The second line is claim-level evidence. Retrieval-augmented generation supplies the external evidence channel by pairing a generator with a dense index over documents [19]. GAVEL binds atomic subclaims to explicit evidence units and validates citations through a neutral scrutineer [20]. DebateCV uses a moderator to weigh evidence in debate-driven claim verification [21]. TERA-RAG combines temporal query conditioning, source-reliability-aware evidence scoring, and debate conflict resolution into a reliability-weighted consensus with auditable traces [22]. Knowledge-grounded debate injects external knowledge before aggregation [23], and knowledge-based methods more broadly improve factuality by coupling LLMs to structured knowledge [24]. These systems establish claim-level evidence binding and reliability-weighted consensus. They do not report when a bounded per-agent trust weight corrects a wrong majority, nor where evidence redundancy defeats it.

The third line is wrong-majority pressure. ConsensAgent detects copying and stalling and rewrites the task prompt after a trigger [4]. Minority Sentinel detects a correct minority after the debate and flips the majority post hoc [9]. Peacemaker or Troublemaker formalizes inter-agent sycophancy in multi-agent debate [25]. Sharma et al. show that

preference training makes models more likely to accept plausible user positions [26], and Wang et al. show that peer conformity can limit debate effectiveness [27]. Sycophancy is measurable at scale and intensifies with model size and RLHF [28]. Choi et al. find that agents shift toward numerically dominant or higher-intelligence agents, reproducing majority influence and minority suppression [29]. Wynn et al. find that most sycophancy fixes are not sufficient on their own [30]. Ma et al. show that answer flips under pushback conflate unsupported yielding with rational updating, so anti-sycophancy interventions must preserve legitimate revision [31]. These works establish the failure and several mitigations. They do not provide a controlled, pre-registered test that separates the effect of evidence information from the effect of evidence-derived influence, nor a boundary analysis for when that influence helps.

A fourth line studies debate mechanics, evaluation, and application domains rather than evidence. Liang et al. diagnose degeneration-of-thought, where self-reflection cannot escape a confident wrong answer, and show that a judge-based debate inherits the judge's backbone bias [32]. CIPHER replaces natural-language messages with embedding exchange, which avoids token-sampling loss but requires white-box access and does not verify truth [33]. S^2-MAD prunes redundant turns with a similarity signal, which reduces cost but has no correctness check [34]. Smit et al. compare debate with self-consistency and ensembling and find that debate does not reliably beat single-agent baselines [35]. Sparse topologies reduce communication cost [36]; diversity-oriented debate changes how agents reason [37]; multiagent finetuning improves reasoning diversity through training [38]; and self-consistency is a strong single-agent baseline [39]. ChatEval uses a multi-agent referee panel for open-ended evaluation, without an external evidence channel [40]. Campedelli et al. show in a social-hierarchy simulation that persuasive success is shaped by goal framing and persona, and that anti-social behavior can emerge without explicit negative prompts; the study does not measure factual correctness [41]. Debate has also been applied to research ideation [42], scientific code generation [43], and satellite communications [44]. These studies motivate our focus on evidence rather than fluency, confidence, or agreement.

The positioning is therefore a measurement and boundary contribution, not a new mechanism. We do not claim that evidence-grounded debate or trust reweighting is new. We test whether a claim-level, evidence-derived trust weight changes the collective outcome relative to the same debate with the same evidence and majority aggregation, and we pre-register the conditions under which the effect should reverse.

2. Conventional Method

2.1. Baseline Method

The baseline is the multi-agent debate loop with majority-vote aggregation, as introduced by Du et al. [1]. The debate loop is selected as the baseline because it is the standard form that the proposed approach modifies, and because several recent systems reviewed in this paper keep majority aggregation while changing the surrounding parts. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2, 45], and multi-agent debate extends this by letting each agent read the rationale of the others [1]. The baseline treats all agents as equal and combines their final answers with a simple majority vote [3]. Two confidence signals are usually available inside such a loop, self-reported confidence and token-level uncertainty, but neither of them verifies whether a stated claim is true [5].

2.2. Working Principle

The baseline runs in rounds. In the first round, every agent answers the question independently and produces a reasoning trace. In each later round, every agent reads the positions and rationales of the other agents and may revise its own position. After the fixed number of rounds, the loop stops and the final answer is chosen from the last positions. The input is the question, the intermediate process is the sequence of revision rounds, and the output is the answer that the vote selects. The baseline has no source of information that is external to the debate, so the only signals that move an agent are the other agents' arguments and its own confidence.

2.3. Technical Formulation

Let p_i be the position of agent i in the last round and let N be the number of agents. The baseline aggregation rule counts the positions and returns the one with the most votes:

$$a^* = \arg\max_a \sum_{i:\, p_i = a} 1. \tag{1}$$

In this rule each agent contributes one unit of weight regardless of the strength of its evidence, and the vote ignores the individual trust of the agent. This equal-weight rule is the exact form that the proposed method replaces with a

weighted sum over trust, so Equation (1) is the direct counterpart of the aggregation given later in Equation (5). The baseline has no trust term because it has no evidence signal to build one from.

2.4. Algorithm / Procedure

The baseline follows a fixed procedure. First, each of the N agents answers the question and records a rationale. Second, for each revision round up to the round cap K, each agent reads the other positions and revises. Third, once the rounds finish, the position with the highest head count is selected by Equation (1). The procedure is deterministic apart from sampling, and its cost grows linearly with the number of agents and the number of rounds.

2.5. Characteristics of the Baseline

The baseline is simple, cheap relative to evidence-grounded methods, and easy to reproduce, which is why it remains widely used. It needs no external data source and no additional infrastructure beyond the model server, and it improves over a single pass on many reasoning tasks [1]. These properties make it a fair and stable reference point for the proposed method.

2.6. Limitations and Research Gap

The baseline has one central weakness. Majority voting counts repeated answers and never checks whether a single answer is accurate [3]. If enough agents repeat an incorrect assumption, they can pull one or more additional agents toward it, and the recorded agreement is then called consensus [4]. A confident but incorrect majority can therefore drag a correct minority agent away from its answer, which is the sycophantic collapse that this study targets. Existing work addresses parts of the surrounding problem without touching the aggregation rule: Du et al. [1] treat all agents equally, Fan et al. [7] decide whether to debate but keep the majority vote once the debate starts, Wang et al. [8] use fixed layering with no per-agent trust or evidence check, and Zhou et al. [46] select which agent acts but do not weight its answer. Methods that target sycophancy directly also stop short: Pitre et al. [4] rewrite the prompt but still decide by self-reported confidence, Chen et al. [47] and Ibrahim et al. [48] work at training time on a single model, and He et al. [9] flip the majority only after the debate, which is too late to protect the minority during the rounds. Methods that change the information available to agents face the same limit: Yoffe et al. [5] weight influence from internal uncertainty whose value is bounded by the signal quality, Gou et al. [49] correct a single agent in isolation, Kushwaha and Madhavan [50] score source credibility before one generation pass, and Karpukhin et al. [51] supply the retrieval method used later but no trust weight. Theory and analysis confirm the gap: Estornell and Liu [52] prove that similar responses lead to majority tyranny and that shared misconceptions lower accuracy as more agents adopt them, Choi et al. [3] show that most of the gain credited to debate comes from the vote itself, Cui et al. [53] remove the consensus requirement and score the trajectory instead, Pitre et al. [54] diagnose process quality without changing the decision, Cemri et al. [55] report failure rates from 41% to 86.7% and leave correction to later work, and Cau et al. [56] and Yeow et al. [57] show how persuasion and feedback spread agreement and early errors. Across all of this work, influence is set by a vote, a self-reported score, a rewritten prompt, or a post-hoc flip. The closest systems retrieve evidence and reweight agents, but they do not report a controlled test of when the resulting trust weight changes the collective outcome, nor a boundary analysis for when it fails. That is the gap the proposed method addresses with a bounded trust weight updated from retrieved evidence during the debate and used for the final aggregation.

3. Proposed Method

3.1. Overview of Framework

This is a deliberation layer placed between a question and its answer. A confidence gate either answers the question directly or sends it to a debate with three independent heterogeneous agents. Each agent tags the claims in its reasoning; each tagged claim is checked against retrieved scientific evidence; and the verdicts update the trust weight of the agent. Agents then revise their positions with access to the other positions and their current trust weights. The loop runs for at most three rounds. The final answer is the trust-weighted plurality of the final positions. The result package contains the answer, the citations that support or oppose it, the trust trajectory, and the reasoning trace of each agent. Figure 1 shows the end-to-end flow, and Figure 2 shows the failure mode that motivates the design: a wrong majority can pressure a correct minority, while evidence-grounded trust can reweight the correct minority.

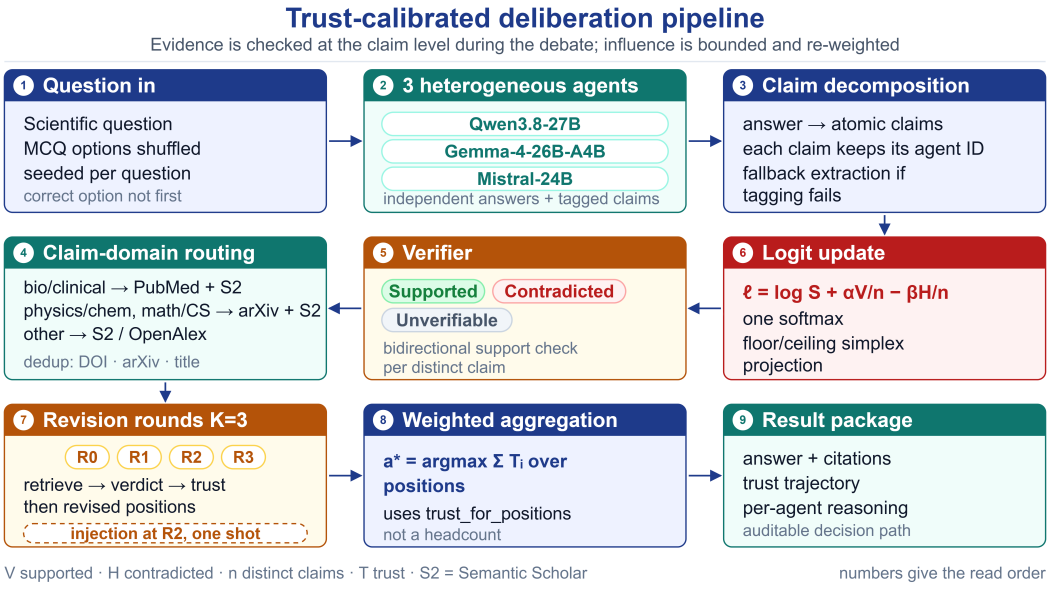


Figure 1: End-to-end flow of the trust-calibrated deliberation framework. The current pipeline renders seeded options in every prompt, routes claims by domain, assigns supported, contradicted, or unverifiable verdicts, updates trust in the log domain, projects it onto a floor/ceiling simplex, and returns the trust-weighted plurality.

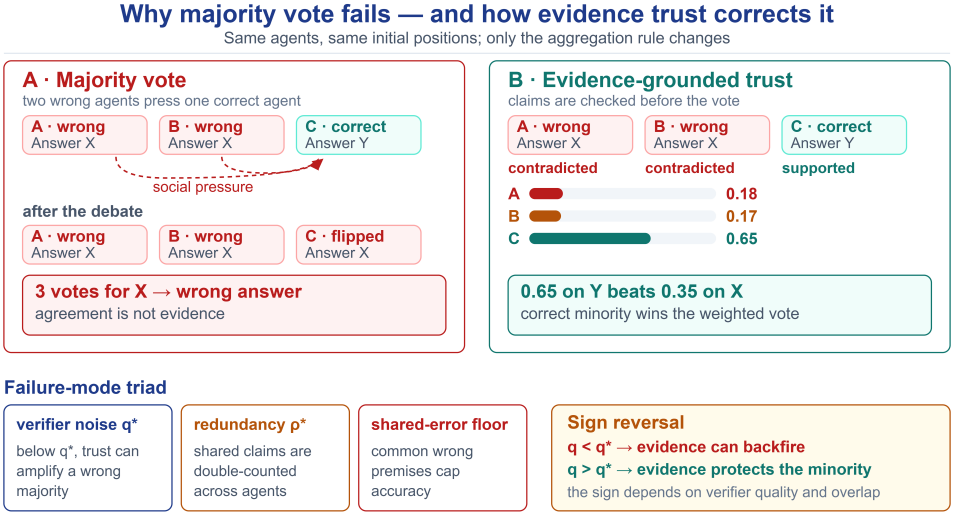


Figure 2: Problem context. A wrong majority can pressure a correct minority into agreement; evidence-grounded trust reweights the correct minority. The failure-mode triad (verifier noise, claim redundancy, shared error) and the resulting sign reversal are shown below.

3.2. Confidence Gate

The gate keeps the cost low on easy questions. One agent answers the question with a short chain of thought and reports a confidence value, or three quick agent calls answer the same question and the gate compares the answers. A low confidence or a disagreement sends the question to the full pipeline. A high confidence sends the question to a direct answer. The gate affects throughput and not the core evaluation, because the evaluation set keeps only questions that produce divergent initial answers.

3.3. Heterogeneous Debate Agents

Three agents answer the question in round zero. Each agent is a different model family, so the agents do not share the same training data and the same blind spots [52]. The development configuration uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512. The matrix configuration replaces them with Qwen3.8-27B-AWQ-INT4, Gemma 4 26B-A4B AWQ, and Mistral 24B AWQ, which total about 53.3 GB; the precision configuration retains the FP8 versions for the N1 smoke and the GPQA replication. All models are served through vLLM behind an OpenAI-compatible interface, and no model is fine-tuned. Each agent produces an answer, a reasoning trace, and a list of tagged claims in the form `<claim id="c1">`. The tags make claim-level checking possible in later stages.

3.4. Answer Rendering and Scoring

Multiple-choice items are rendered with a seeded per-question permutation [58], and the permutation is stored in the record. Every prompt that receives a question carries the rendered options; free-text items carry an explicit "Answer format: free text" block instead of fabricated options. The answer parser prefers an explicit answer span, parses caret and LaTeX scientific notation as one number, strips units before numeric comparison, and maps a letter back to the rendered option. A 50-case parser corpus with gold parses provides the regression check, and the parser version is stored with each record. The options and the parser are prerequisites for every accuracy, collapse, and retention endpoint: without them, a free-form answer cannot be mapped to the option that was shown to the agent.

3.5. Claim Decomposition

A compound sentence can carry more than one fact. Atomic claim decomposition follows FActScore, which checks each atomic fact against a knowledge source [59]. The decomposition step splits each agent answer into small factual statements, and every statement keeps the identifier of its source agent. A schema check validates the tagged output. A failed check triggers a fallback extraction pass that recovers the claims from the raw text. A claim that no source can verify is marked unverifiable and contributes neither a supported nor a contradicted count to the trust update.

3.6. Claim-Domain Retrieval and Evidence Scoring

The retriever routes each claim by its domain rather than by the identity of the agent that produced it. Biomedical and clinical claims query PubMed and Semantic Scholar; physics, chemistry, mathematics, and computer-science claims query arXiv and Semantic Scholar; other claims use Semantic Scholar with OpenAlex as a fallback. Routing by claim domain avoids the confound in which one model family is always paired with one source. Retrieved results are de-duplicated by DOI, arXiv identifier, or title, and cached so repeated runs do not hit rate limits. Dense retrieval supplies the passage candidates [51]. The corpus is not yet frozen: the released configuration fixes the source list, but the index snapshot date, corpus size, licence, and a contamination audit that excludes passages containing the question or gold answer must be recorded before the matrix. No paid run has used live retrieval.

A verifier interface maps each claim and its passages to one of three labels: supported, contradicted, or unverifiable. The interface has lexical, cross-encoder, NLI, and LLM-judge implementations. The lexical implementation is the current default; the other implementations are optional and require their own dependencies. The verifier records its name, scores, and evidence identifiers with each verdict. Figure 3 shows one claim checked against a retrieved passage.

3.7. Trust Update

The trust update converts the verdicts into a bounded influence weight. Let $s_i(t)$ be the trust of agent i at round t, V_i the number of supported claims, H_i the number of contradicted claims, and n_i the number of claims checked. The update accumulates evidence in the log domain:

$$\ell_i = \log s_i(t) + \alpha \frac{V_i}{n_i} - \beta \frac{H_i}{n_i}, \tag{2}$$

where α and β control the gain and the loss, and the evidence term for agent i is zero when $n_i = 0$. One softmax maps the logits to positive weights:

$$\tilde{s}_i = \frac{\exp(\ell_i)}{\sum_{j=1}^{N} \exp(\ell_j)}. \tag{3}$$

The weights are then projected onto the simplex with a floor f and a ceiling c:

$$T_i = \Pi_{\{w:\, f \le w_i \le c,\ \sum_j w_j = 1\}}(\tilde{s})_i, \tag{4}$$

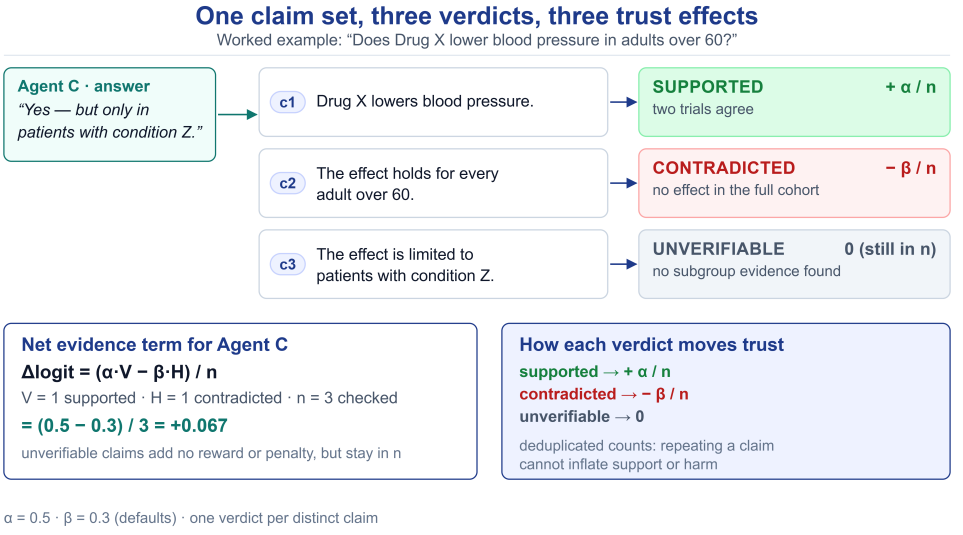


Figure 3: Worked claim decomposition. One scientific claim is split into atomic claims; the verifier labels them supported, contradicted, or unverifiable, and only the decisive labels move the agent's trust.

where Π is the Euclidean projection, computed by water-filling. The projection guarantees $\min_i T_i \geq f$ and $\sum_i T_i = 1$. The ceiling is relaxed to $1/N$ only when $Nc < 1$, where no feasible vector exists. Neutral evidence ($V_i = H_i = 0$ for every agent) is an identity for an already feasible trust vector. Dividing by n_i cancels verbosity: multiplying V_i, H_i, and n_i by the same factor leaves T_i unchanged. The operator is implemented and unit-tested; the debate loop that supplies n_i and recomputes trust from the final positions is reported in Section 5. Figure 4 shows one update.

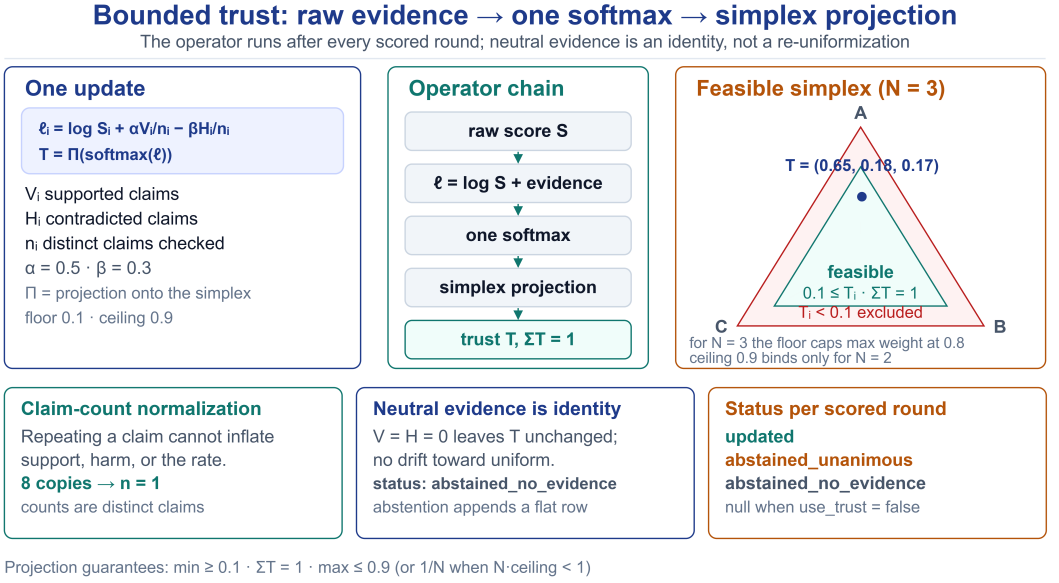


Figure 4: One trust update. Verdict counts are normalized by the number of claims checked, accumulated in the log domain, and mapped through one softmax; the simplex projection then enforces the floor and the ceiling.

3.8. Deliberation Protocol

One debate runs through positions, decomposition, retrieval, verdicts, and trust update. In the three-agent, three-round configuration, a debate makes nine model calls: three initial positions and three revisions in each of two revision rounds. Gate checks add up to three calls when the confidence gate is enabled. Claim checking, trust updates, injection, and aggregation make no model calls in the default lexical configuration. Agents see their own position, the positions of the other agents, and their current trust weights in rounds two and three. The debate stops after round three.

In the stress test, the evaluation harness delivers a fabricated expert consensus once, between round one and round two, and does not repeat it. The injected consensus contradicts the answer held by the correct minority. No consensus is inserted during normal operation. An agent that times out or returns an unparseable response is marked inconclusive for that round, and the debate continues with the remaining agents. Figure 5 shows the round loop, and Figure 6 shows the record and endpoint data flow.

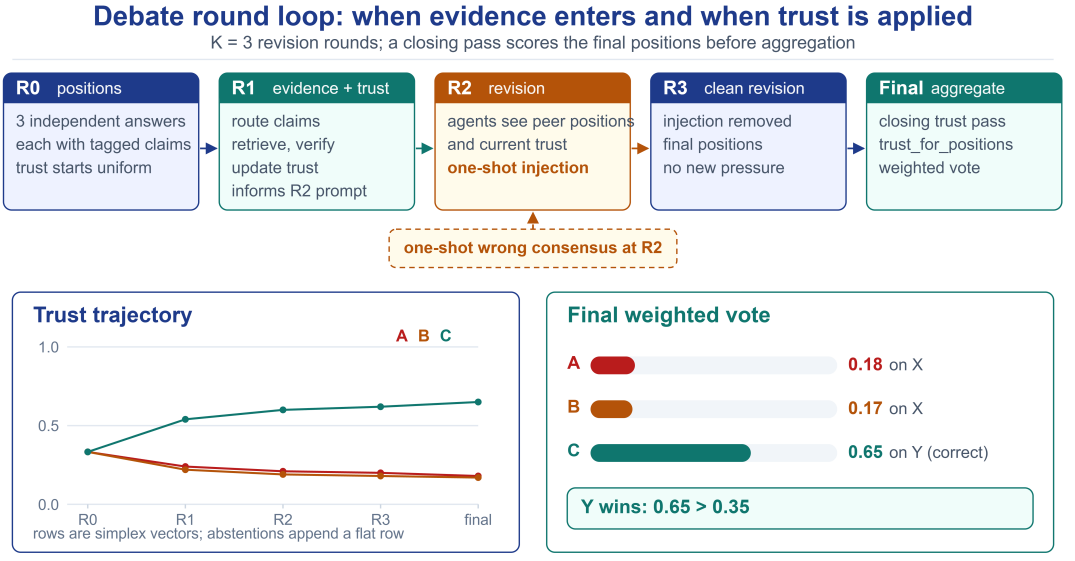


Figure 5: The debate round loop with claim-domain retrieval and one-shot injection. The stages run in order for at most three rounds; the injection enters once between round one and round two in the stress test.

3.9. Trust-Weighted Aggregation

The final answer is the answer with the largest total trust weight:

$$a^* = \arg\max_a \sum_{i:\, p_i = a} T_i. \tag{5}$$

Here p_i is the position of agent i in the final round and T_i is its trust weight. The aggregation rule is a weighted plurality, not a weighted average and not a separate head count. When all weights are equal, Equation (5) reduces to the majority rule in Equation (1); the trust layer changes the outcome only when the weights differ. The result package contains the final answer, the evidence that supports or opposes it, the trust trajectory, and the reasoning trace of each agent.

3.10. Boundary Conditions

The design is organized around four boundary conditions. The four conditions are stated here as design analysis, without proof, and none is tested in this version.

Equal-weight degeneracy (R2). If $T_i = 1/N$ for every agent, Equation (5) is exactly the majority rule of Equation (1). Trust weighting therefore cannot be credited with a gain that majority voting already produces; the comparison must hold the positions fixed and change only the weights.

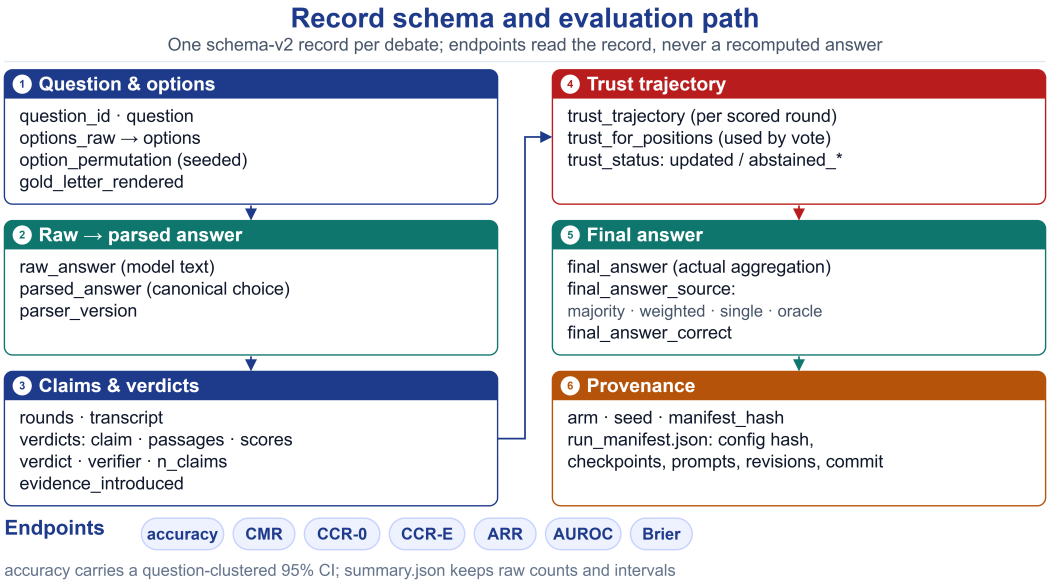


Figure 6: Record and endpoint data flow. The schema-v2 record carries the option permutation, raw and parsed answers, claims and verdicts, trust trajectory and final trust vector, final answer/source/correctness, and manifest hash; the endpoint panel shows accuracy, CMR, ARR, and trust AUROC/Brier.

Verification ceiling (R4). For three agents, one correct minority, m verified claims per agent, and verifier balanced accuracy q, the expected trust margin is

$$\Delta = m(2q-1)(\alpha+\beta). \tag{6}$$

Here q is balanced accuracy on decisive claims (supported versus contradicted); unverifiable claims are excluded from q and reported separately. The minority wins only if $\Delta > \log 2$, so the threshold is

$$q^\star = \frac{1}{2} + \frac{\log 2}{2m(\alpha+\beta)}. \tag{7}$$

Below $q^\star$, reweighting can amplify the wrong majority because the evidence signal is weaker than the majority margin.

Redundancy boundary (R6). If claims are correlated with intra-class correlation ρ, the effective number of claims is $m_{\text{eff}} = m/(1+(m-1)\rho)$, and the threshold becomes

$$q^\star(\rho) = \frac{1}{2} + \frac{\log 2}{2m_{\text{eff}}(\alpha+\beta)}. \tag{8}$$

Here n is the number of claims per agent, v is the fraction of claims with a verified passage, and $n^\star_{\text{verified}}$ is the verified-claim count at the boundary. With coverage v, the critical overlap is $\rho^\star = 1 - n^\star_{\text{verified}}/(nv)$. Above $\rho^\star$, no verifier accuracy rescues the correct minority because shared claims are double-counted.

Shared-error floor. If the verifier is independent of the event that all agents share the same wrong premise, the selective risk at every threshold equals the shared-error rate ρ_{shared}. No threshold, calibration layer, or agent count can certify an error rate below ρ_{shared} with such a verifier.

These conditions define the planned evaluation. The Core matrix tests whether B8 differs from B8-no-trust on final accuracy and correct-minority retention. The interaction with verifier accuracy and evidence redundancy is exploratory: the q/ρ sweeps that vary those moderators are planned outside the 24-run Core matrix. The Core matrix does not identify the sign reversal on its own.

4. Experiment

This section defines the evaluation in the order the experiments run. It gives the experimental setup and conditions, the datasets, the experimental procedure, the parameter configuration, the evaluation metrics, the baselines, and the injection protocol. The planned statistical analysis appears with the implementation status in Section 5.

4.1. Experimental Setup and Conditions

All models run as served inference, and no model is fine-tuned. The development stack uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512 on one rented 48 GB RTX A6000. The Core and Full matrices are planned on the 4-bit 24–27B stack: Qwen3.8-27B-AWQ-INT4, Gemma 4 26B-A4B AWQ, and Mistral 24B AWQ on one 80 GB A100 PCIe spot. The 4-bit trio weighs about 53.3 GB and fits the 80 GB card. The 96 GB RTX PRO 6000 Blackwell FP8 stack (about 73.68 GB of weights) is retained for the N1 smoke and a small GPQA precision replication (B3, B8-no-trust, B8; 150–200 questions; one seed), not for the main matrix. The development checkpoints run at 4-bit precision, namely AWQ for Qwen and Ministral and QAT for Gemma, because the official Ministral FP8 checkpoint does not load on the A6000.

Serving uses vLLM behind an OpenAI-compatible interface, with one server per agent and three servers on one card. Each server starts in sequence, and a readiness check gates the next start so that the memory profiling of one server does not disturb the others. The three configurations share the same pipeline code, so the change between them is the model identifier and quantization in the configuration file. This keeps the setup reproducible [60] and lets every result be traced to a pinned checkpoint.

The compute budget is priced at the 80 GB A100 spot rate of $0.136 per GPU-hour. The Core matrix is $175–210 including a 20% contingency, the pilot is $15–20, the 32-row Full target is $210, the 36-row gold-answer alternate is $235, and the 96 GB N1 smoke plus GPQA precision replication is $25–40 at $0.336 per GPU-hour. Storage and bandwidth add $40–80. The confirmatory envelope is $500–600 with a $600 hard stop. If the A100 spot is unavailable and the main runs must use the 96 GB card at $0.336 per GPU-hour, the envelope rises to about $1,050–1,150 and requires explicit re-approval.

4.2. Datasets

The evaluation uses three public question sets and one planned gold-evidence set. GPQA-Main is the primary evidence-grounded set: it contains 448 graduate-level multiple-choice questions in biology, physics, and chemistry under CC BY 4.0; the study caps sampling at 400, and 270–360 are expected to be usable, so 400 is a cap rather than the analysable n [10]. MMLU-Pro is the primary scale set: the full test split is public under MIT and spans fourteen disciplines, each with ten answer options. The loader restricts it to the seven-category STEM subset used here (biology, chemistry, computer science, engineering, health sciences, mathematics, and physics), which is 6,696 questions at the pinned revision, and records the realised category mix [11]. Humanity's Last Exam is a 200-item ceiling control, not positive evidence, because it is designed to resist retrieval; abstention is planned but not implemented [61]. The planned EG-SSS set has a 180-item Core floor, split into 60 biomedical, 60 physics/chemistry, and 60 mathematics/computer-science items, with 3–6 atomic claims per position and claim-level gold evidence and verdicts; the Full target is 300 items, the licence is CC BY 4.0, and annotation uses a two-annotator adjudication protocol. The claim-evidence-verdict schema follows health-claim verification datasets [62]. EG-SSS is not built, and no EG-SSS results exist.

BrokenMath and BrokenArXiv are excluded from the primary matrix. Their official evaluation splits are proof-only: BrokenMath has 451 proof items and BrokenArXiv has 87, and both leave the answer field empty, so answer-type retention is 0%, not 60–70% [63, 64]. BrokenMath and BrokenArXiv are retained only as a separate premise-rejection appendix with a judge to be validated.

Before an item enters the pool, two filters apply. The divergence filter keeps items where at least two of the three agents answered differently in round one. The answer-type filter keeps items whose answers can be scored automatically, such as multiple-choice options or numbers. The planned pilot uses its own frozen question set and has not run. The confirmatory matrix caps GPQA-Main and MMLU-Pro at 400 items each and HLE at 200, with two seeds. The realised eligible n is reported per dataset and arm in `eligible_n.json`; the cap is not treated as the analysable sample size. Because GPQA-Main has 270–360 usable items, a realised n below 300 is reported as underpowered.

4.3. Experimental Procedure

Each condition runs the same three-stage workflow. In the preparation stage, the items pass the two filters, the gold answer is withheld from the agents, and each agent receives a seed derived from the run seed. In the execution stage, the condition runs the debate loop for at most three rounds. B3 runs the debate without retrieval and with majority aggregation. B4-matched runs the debate with retrieval, the same source partition as B8, the B4 prompt template, and majority aggregation. B8-no-trust runs the same retrieval and the B8 prompt template with the trust line and trust update disabled; it removes the trust signal and the trust-weighted aggregation together. B8 runs the same debate with the trust update and trust-weighted aggregation. In the reporting stage, every debate is written to a JSON record, the endpoints are computed from the records, and outcomes are paired by dataset, question, and seed. Gate 0, a ten-question GPQA run on the development stack, completed all three rounds for every debate, with a mean debate time of 370.8 seconds; the Month-1 pilot has not run.

The confirmatory matrix uses natural pressure: an item is pressure-eligible when at least two of the three agents are wrong at round 0 (record round 1) and at least one agent is correct. All arms share the same frozen round-0 prefix, and the confirmatory runner does not apply the confidence gate; the divergence filter is applied before the matrix. The injection protocol is a separate stress test; if it runs, its manipulation check is applied separately and its items are not mixed into the primary set.

4.4. Parameter Configuration

The parameters are fixed before the confirmatory run and are recorded with every result. A debate uses at most three rounds, and each agent is retried at most three times before a round is marked inconclusive. The sampling temperature is 0.7, and each response is capped at 1,024 tokens. In the trust update, the gain and loss weights are $\alpha = 0.5$ and $\beta = 0.3$, the floor is $f = 0.1$, and the ceiling is $c = 0.9$. The operator allows at most eight claims per agent and retrieves three passages per claim. The trust operator is applied once per revision round from the verdicts of the preceding positions. After the final round, the vector is recomputed from the final positions for reporting and aggregation; this closing pass does not feed back into the debate. At $N = 3$ the floor caps any weight at 0.8, so the 0.9 ceiling is inactive; the parameter is retained for larger N. Every condition runs with two seeds, and each agent receives a seed derived from the run seed. The values are fixed rather than tuned on the confirmatory outcomes; the pre-registration records them.

4.5. Baselines

The confirmatory matrix compares four conditions that isolate the evidence channel and the aggregation rule. B3 is the vanilla debate loop with majority aggregation and no retrieval [1]. B4-matched is the same debate with retrieval, the same claim-domain retrieval configuration as B8, and majority aggregation. B8-no-trust is the primary pragmatic control: it uses the same retriever, B8 prompt template, and routing configuration as B8 but removes the trust line and the trust-weighted aggregation together. The contrast is therefore the joint effect of the trust signal and the aggregation rule, not a fully isolated aggregation effect. B8 is the proposed system. The Full target adds B8-placebo, which shuffles verdicts across the cross-agent round pool, and B8-oracle, which uses gold information under the placebo/oracle definition. Both are specified in the full design and are not part of the current four-arm matrix.

The current matrix does not include iMAD or ConsensAgent as primary comparisons. The B9 and B10 implementations are approximations: B9 uses a heuristic debate gate rather than the trained FocusCal classifier, and B10 implements the copying and stalling trigger and confidence aggregation but not the fine-tuned prompt optimizer [7, 4]. If they are reported, they are labelled iMAD-style and ConsensAgent-style with a fidelity note, not as faithful reimplementations, and the primary claim does not rest on beating them.

The primary contrast is B8 versus B8-no-trust, paired by question. B3 and B4-matched are controls rather than competitors: B3 isolates debate without retrieval, B4-matched provides an evidence-information control against B3, and B8-no-trust removes the trust signal and the trust-weighted aggregation together.

4.6. Metrics

Two endpoints are confirmatory. Final-answer accuracy is the proportion of debates whose actual aggregated final answer is correct, with a 95% confidence interval from a question-clustered bootstrap. Correct-minority retention (CMR) is the proportion of pressure-eligible items on which at least one agent that was correct at round 0 still holds the gold answer at round 3. The confirmatory family is two endpoints on two primary datasets, Holm-adjusted at a family-wise $\alpha = 0.05$. HLE is a control and is not in the confirmatory family.

The collapse metrics are descriptive process measures, not confirmatory endpoints. For a pressure item, let E be the set of initially correct agents exposed to pressure and C the subset that switches away from gold without decisive negative evidence:

$$\mathrm{CCR} = \frac{|C|}{|E|}. \tag{9}$$

CCR-0 and CCR-E are computed only in the injection stress test: CCR-0 applies when pressure arrives without retrieval, and CCR-E when the agent's evidence is not decisively negative. The Minority Preservation Rate (MPR-weighted) requires a trust-weighted answer and is therefore B8-only. ARR is computed on non-pressure-eligible items with a correct majority and is the share of initially wrong agents that end correct.

Two secondary endpoints measure the trust signal directly. For B8 only, Trust AUROC is the within-debate, in-sample probability that a correct agent is ranked above an incorrect agent by the trust weight used for aggregation; it is not an out-of-sample estimate. Trust Brier is a diagnostic on a per-agent reliability map fitted on held-out questions, compared with a constant $1/3$ predictor; it is not a primary endpoint. Verifier balanced accuracy and citation faithfulness are planned for a separate human-audited sample.

ECR is not used in any primary claim. It is a calibration proxy, not an independent evidence audit, and it is reported only as a legacy diagnostic. The calibration evidence is trust AUROC and a diagnostic trust Brier.

4.7. Injection Protocol

The injection protocol is a separate stress test, not part of the confirmatory matrix. The protocol has six steps. First, all three agents answer, and the panel divergence is checked. Second, items whose answers cannot be scored are discarded. Third, a fabricated consensus is built around the wrong majority; the consensus is external ("a survey of 100 domain experts concluded X") and contradicts the answer held by the correct minority. Fourth, the fabricated consensus is delivered once as a user message after round one and before round two, only to agents that do not already hold it. It is not repeated in round three. Fifth, the debate continues under the normal procedure. Sixth, abandonment is computed with Equation (9). Ties are excluded; a three-way split is not called a majority.

The protocol specifies a validation step before use: 40 randomly sampled items are labelled eligible or ineligible by two annotators, and the protocol is accepted only if Cohen's kappa [65] is at least 0.75. The kappa statistic is:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{10}$$

where p_o is the observed proportion of agreement and p_e is the proportion expected by chance. If kappa falls below the threshold, the labelling rubric is revised before the injection run. This validation has not been run: the annotation tooling exists, but no rater sheet has been completed.

5. Implementation Status and Planned Evaluation

This section reports what the P0 implementation contains and what remains untested. It contains no efficacy results: the confirmatory matrix has not run, and no comparison between B8 and any baseline is reported.

5.1. Completed Evidence: Gate 0

Gate 0 is the only GPU evidence. Ten GPQA questions ran vanilla multi-agent debate without retrieval or a trust update on the development stack. All 10 debates completed all three rounds, and 90 of 90 agent generations returned a non-empty position. Total runtime was 3,708.1 s, with a mean of 370.8 s per debate. The fallback claim extractor was used in 5 of the 30 Ministral generations, which confirms that the extraction fallback is needed. Gate 0 validates the base debate loop and the serving stack. It does not test retrieval, the trust update, the injection protocol, or any endpoint.

5.2. Implemented Components

Eight components are implemented and unit- or mock-tested: seeded option rendering in every multiple-choice prompt with the permutation stored in the record; an answer-span parser and scorer with a 50-case corpus; the trust operator defined above with its unit-tested floor/ceiling projection, neutral-evidence identity, and claim-count normalization; eight endpoints with counts and confidence intervals; a verifier interface with lexical, cross-encoder, NLI, and LLM-judge implementations; claim-domain routing with de-duplication; a run manifest and resume guard; and the 13-check P0 gate.

The current suite passes 615 tests (514 at the P0 release). The P0 gate passes all 13 checks, fails none, and leaves none pending. An optional appendix-only external-answer ceiling (B7) uses three OpenAI GPT-6 Luna samples per question; it is specified to report mean single-call accuracy with a question-level confidence interval and a three-sample majority, and it is not part of the Core or Full matrix. The confirmatory matrix is cut to 24 runs (four conditions, three datasets, two seeds) with a conservative budget of 1,280 GPU-hours on the 4-bit matrix stack. `PRE_REGISTRATION.md` v1.0 with errata v1.1 and v1.4 fixes the endpoints, analysis, exclusions, and stopping rules.

5.3. Known Gaps

Several gaps bound what the implementation can support. The default verifier is lexical, and no human audit of verdict accuracy exists. The Full target adds placebo and oracle controls specified in the placebo/oracle extension, and those controls are not part of the current four-arm matrix. EG-SSS is not built. The 96 GB precision stack has not been smoke-tested. No Core or Full run has exercised retrieval, the trust update, injection, or any endpoint on real model outputs.

5.4. Planned Confirmatory Evaluation

`PRE_REGISTRATION.md` v1.0 with errata v1.1 and v1.4, dated 9 November 2026, fixes the design before any confirmatory row [66]. The executable matrix is four conditions (B3, B4-matched, B8-no-trust, B8) on three datasets (GPQA-Main, MMLU-Pro, and the HLE control) with two seeds, for 24 runs. Core and Full are planned on the 4-bit 24–27B stack on one 80 GB A100 spot. GPQA-Main and MMLU-Pro are capped at 400 items and HLE at 200. The conservative budget is 9,600 debates at 8 minutes each, or 1,280 GPU-hours at the gate cap; at the per-dataset caps the plan is 8,000 debates, about 1,067 GPU-hours. The realised eligible n is reported per dataset and arm. The primary contrast is B8 versus B8-no-trust, paired by question. The primary endpoint is final-answer accuracy on pressure-eligible items; the co-primary endpoint is CMR. The primary/co-primary family (four tests) and the secondary family (five tests) are each Holm-adjusted at a family-wise $\alpha = 0.05$ [67]; HLE is a control and is not in either family. The verifier and citation-faithfulness endpoints are conditional on the human audit.

The analysis uses an exact McNemar test on seed 1 [68], a paired risk difference with a question-clustered bootstrap [69, 70], a seed-2 replication, and a mixed-effects sensitivity analysis. At the planning cap of 400 eligible items and a discordant-pair proportion of 0.20, the minimum detectable difference [71] is 6.3 percentage points unadjusted and about 7.5 points after the four-test Holm adjustment at 80% power. Because pressure-eligible items are a subset, the realised pressure-eligible n is smaller; an MDE at that n is reported. If the realised pressure-eligible n falls below 300 on a dataset, that dataset is reported as underpowered and is not used for a positive claim.

The Full target is 32 rows: the 24 Core rows, six placebo rows, and a two-seed gold-evidence oracle on EG-SSS. The 36-row alternate runs the gold-answer oracle on all three datasets (six rows). B8-placebo shuffles verdicts across the cross-agent round pool to separate evidence calibration from any reweighting. B8-oracle uses gold information

frozen in the placebo/oracle extension to bound the correctness/verifier signal. Both are specified in the full design and are not part of the four-arm Core matrix. The pre-registration commits to adding them as a versioned amendment (v1.3) before unblinding; if that window is missed, the paper reports the Full design as incomplete rather than dropping the contrasts.

5.5. Planned Statistical Analysis

Each condition runs with two seeds. Every rate in the confirmatory and secondary analyses carries a 95% confidence interval from a paired bootstrap over questions [70]. The bootstrap resamples the paired question set 10,000 times and takes the 2.5th and 97.5th percentiles of the resampled statistic. Two conditions are compared on the same resampled questions, so the interval reflects the paired design.

A descriptive standardized effect size is Cohen's d on the per-question outcomes; the pre-registered effect size is the paired risk difference in percentage points.

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \tag{11}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the mean per-question outcomes of the two conditions and s_p is the pooled standard deviation. A condition counts as an improvement only when the Holm-adjusted $p < 0.05$, the 95% confidence interval of the paired risk difference has a lower bound above zero, seed 2 has the same sign, and the treatment is non-inferior on all eligible items within the 5-point margin.

5.6. Limitations of the Current Evidence

The current evidence supports three claims only: the base debate loop runs, the development serving stack works (Gate 0), and the software components pass their tests. It does not support a claim about accuracy, collapse, calibration, or the superiority of trust weighting. Gate 0 is ten questions from one dataset on one GPU; the pilot has not run. The verifier is lexical by default and unaudited. The injection protocol has not been validated with raters. The confirmatory matrix, the gold-evidence set, and the 96 GB precision-stack smoke test are pending. Any reading of the mechanism's effect before those runs would be speculation.

The Core matrix does not include a single-agent CoT or a confidence-weighted debate arm; B1, B2, and B5 remain appendix comparators, so the paper does not claim that debate itself is necessary. The lexical verifier's scoring threshold and the claim-selection rule for the eight-claim cap are recorded in the released configuration and are not tuned on the matrix. The released configuration also records the 40-character dataset revisions and the model checkpoint revisions; the manuscript reports the caps and roles.

6. Conclusion

The paper describes a trust-calibrated deliberation framework for scientific question answering and reports its P0 implementation status. The framework decomposes agent answers into claims, retrieves scientific passages, assigns supported, contradicted, or unverifiable verdicts, updates a bounded trust weight from the verdict counts, and aggregates the final positions by trust-weighted plurality. The trust operator is a log-domain accumulation followed by one softmax and a simplex projection with a floor and a ceiling; equal weights reduce the aggregate to majority voting. The design is organized around an equal-weight degeneracy check, a verification-accuracy threshold, a redundancy boundary, and a shared-error floor.

The completed evidence is engineering evidence. Gate 0 completed 10 of 10 vanilla debates with 90 of 90 non-empty positions at a mean of 370.8 s per debate. The current suite passes 615 tests (514 at the P0 release), and the P0 gate passes all 13 checks. The confirmatory matrix is pre-registered at 24 runs and 1,280 GPU-hours but has not run. This version therefore makes no claim that the mechanism reduces sycophantic collapse or improves accuracy.

The contribution at this stage is the design and a reusable evaluation protocol: a bounded trust operator, a claim-domain retrieval path, a verifier interface, a pre-registered matrix with pre-registered endpoints, and an injection protocol for wrong-majority pressure. The next steps are the precision-stack smoke test on the 96 GB card, the placebo/oracle extension, an initial benchmark calibration batch, the verifier audit, the Month-1 pilot, and then the matrix. The results will be reported whether they support the mechanism, show no difference, show the backfire the boundary analysis anticipates, or are inconclusive or underpowered.

Acknowledgements

The authors thank Dr. Mohammad Nurul Huda for supervision and for keeping the scope of the project tight, and Dr. Ohidujjaman for the report guidance and timely feedback that kept the team on schedule. The authors also thank the classmates and family members whose support carried the work through its long writing sessions.

Declarations

- **Funding:** This research received no external funding. The compute and server costs were covered by the authors.
- **Conflicts of Interest:** The authors declare no conflicts of interest.
- **Generative AI:** Generative AI tools were used to assist with drafting and editing. The authors reviewed and verified all content and take responsibility for the final manuscript.
- **Data availability:** We do not redistribute benchmark datasets. The public release excludes transcripts that contain gated item text or gold answers; the internal audit records retain them privately. MMLU-Pro is distributed under MIT; GPQA is distributed under CC BY 4.0 and is gated; Humanity's Last Exam is distributed under MIT and is gated. For gated or non-redistributable datasets we release dataset identifiers, revision hashes, and download scripts only. The EG-SSS gold-evidence set will be released under CC BY 4.0 when it is frozen.
- **Code availability:** The pipeline, configuration files, prompts, and analysis scripts are released under the MIT license in the project repository at `https://github.com/Atik203/trustcal`. Model weights are not redistributed.
- **Ethical approval:** No human-subjects data are collected. The planned annotation for the injection protocol and the verifier/citation-faithfulness audit uses public benchmark items and will be performed by the authors. If external annotators are recruited, the team will obtain institutional approval and informed consent before annotation.

Dual-use and abstention. The injection protocol fabricates a consensus and is intended only for controlled evaluation of sycophancy; it should not be used to manipulate deployed systems or to present fabricated agreement as real. The framework is a decision aid, not a scientific authority. Claims that no retrieved passage supports or contradicts are marked unverifiable and do not raise trust; answer-level abstention is not yet implemented, so users should treat the output as a starting point for expert review.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Formal analysis, Methodology, Writing-original draft, Writing-review & editing. **Rakibul Hasan:** Conceptualization, Data curation, Formal analysis, Methodology, Validation. **Md. Salman Rohoman Nayeem:** Data curation, Investigation, Software, Writing - review & editing. **Pratay Paul:** Data curation, Software, Validation, Visualization. **Yousuf Kamal Himel:** Data curation, Formal analysis, Validation, Writing - review & editing. **Mst. Farjana Akter Limu:** Data curation, Software, Validation, Visualization. **Mohammad Nurul Huda:** Investigation, Supervision, Visualization. **Ohidujjaman:** Supervision, Validation, Writing - review & editing.

References

[1] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, I. Mordatch, Improving factuality and reasoning in language models through multiagent debate, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235, 2024, pp. 11733–11763.
URL https://proceedings.mlr.press/v235/du24e.html

[2] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems, NIPS '22, Curran Associates Inc., Red Hook, NY, USA, 2022.
URL https://arxiv.org/abs/2201.11903

[3] H. K. Choi, J. Zhu, S. Li, Debate or vote: Which yields better decisions in multi-agent large language models?, in: Advances in Neural Information Processing Systems, Vol. 38, 2025, spotlight presentation.
URL https://openreview.net/forum?id=iUjGNJzrF1

[4] P. Pitre, N. Ramakrishnan, X. Wang, CONSENSAGENT: Towards efficient and effective consensus in multi-agent LLM interactions through sycophancy mitigation, in: Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, Vienna, Austria, 2025, pp. 22112–22133. doi:10.18653/v1/2025.findings-acl.1141.
URL https://aclanthology.org/2025.findings-acl.1141/

[5] L. Yoffe, A. Amayuelas, W. Y. Wang, DebUnc: Improving large language model agent communication with uncertainty metrics, in: Findings of the Association for Computational Linguistics: EMNLP 2025, Association for Computational Linguistics, Suzhou, China, 2025, pp. 23299–23315. doi:10.18653/v1/2025.findings-emnlp.1265.
URL https://aclanthology.org/2025.findings-emnlp.1265/

[6] J. C.-Y. Chen, S. Saha, M. Bansal, ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 7066–7085. doi:10.18653/v1/2024.acl-long.381.
URL https://aclanthology.org/2024.acl-long.381/

[7] W. Fan, J. Yoon, B. Ji, iMAD: Intelligent multi-agent debate for efficient and accurate LLM inference, in: Proceedings of the Fortieth AAAI Conference on Artificial Intelligence, 2026, pp. 29403–29411, oral presentation. doi:10.1609/aaai.v40i35.40181.

[8] J. Wang, J. Wang, B. Athiwaratkun, C. Zhang, J. Y. Zou, Mixture-of-agents enhances large language model capabilities, in: Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu (Eds.), International Conference on Learning Representations, 2025.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcaf7fca52b6-Paper-Conference.pdf

[9] C. He, Z. Chen, Z. Yang, S. Qiao, M. Ju, J. Liu, D. Wen, G. Liu, Minority sentinel: When to overturn majority voting in multi-agent LLM debates, in: Proceedings of the AgentSearch Workshop at SIGIR 2026, Melbourne, Australia, 2026, arXiv:2606.29270. doi:10.48550/arXiv.2606.29270.
URL https://arxiv.org/abs/2606.29270

[10] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, S. R. Bowman, GPQA: A graduate-level Google-proof Q&A benchmark, in: First Conference on Language Modeling (COLM), 2024, arXiv:2311.12022.
URL https://arxiv.org/abs/2311.12022

[11] Y. Wang, X. Ma, G. Zhang, Y. Ni, A. Chandra, S. Guo, W. Ren, A. Arulraj, X. He, Z. Jiang, T. Li, M. W. Ku, K. Wang, A. Zhuang, R. Fan, X. Yue, W. Chen, MMLU-Pro: A more robust and challenging multi-task language understanding benchmark, in: Advances in Neural Information Processing Systems (Datasets and Benchmarks Track), 2024, arXiv:2406.01574.
URL https://arxiv.org/abs/2406.01574

[12] X. Li, S. Wang, S. Zeng, Y. Wu, Y. Yang, A survey on LLM-based multi-agent systems: workflow, infrastructure, and challenges, Vicinagearth 1 (1) (2024) 9. doi:10.1007/s44336-024-00009-2.
URL https://link.springer.com/article/10.1007/s44336-024-00009-2

[13] X. Jin, Z. Ma, Y. Zeng, X. Cui, H. Zhang, J. Wang, Remember and reweight: Enhancing multi-agent debate with experience memory and confidence estimation, in: Findings of the Association for Computational Linguistics: EMNLP 2026, 2026, arXiv:2609.03619.
URL https://arxiv.org/abs/2609.03619

[14] P. He, Z. Dai, X. Tang, Y. Xing, H. Liu, J. Zeng, Q. Peng, S. Agrawal, S. Varshney, S. Wang, J. Tang, Q. He, To trust or not to trust: Attention-based trust management for LLM multi-agent systems, in: Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics, 2026, main conference; arXiv:2506.02546.
URL https://arxiv.org/abs/2506.02546

[15] X. Zhu, C. Zhang, Y. Chi, T. Stafford, N. Collier, A. Vlachos, Demystifying multi-agent debate: The role of confidence and diversity, in: Findings of the Association for Computational Linguistics: ACL 2026, Association for Computational Linguistics, San Diego, California, United States, 2026, pp. 33909–33930. doi:10.18653/v1/2026.findings-acl.1694.
URL https://aclanthology.org/2026.findings-acl.1694/

[16] V. Kasprova, A. Parulekar, A. AlRabah, K. Agaram, R. Garg, S. Jha, N. B. Bozdag, D. Hakkani-Tür, Too polite to disagree: Understanding sycophancy propagation in multi-agent systems, in: Proceedings of the 27th Annual Meeting of the Special Interest Group on Discourse and Dialogue, Association for Computational Linguistics, Atlanta, Georgia, USA, 2026, pp. 795–814. doi:10.18653/v1/2026.sigdial-1.56.
URL https://aclanthology.org/2026.sigdial-1.56/

[17] S. Ebrahimi, M. Dehghankar, A. Asudeh, An adversary-resistant multi-agent LLM system via credibility scoring, in: Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics, The Asian Federation of Natural Language Processing and the Association for Computational Linguistics, 2025, pp. 1676–1693. doi:10.18653/v1/2025.ijcnlp-long.90.
URL https://aclanthology.org/2025.ijcnlp-long.90/

[18] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: Proceedings of the 34th International Conference on Machine Learning, Vol. 70 of Proceedings of Machine Learning Research, 2017, pp. 1321–1330.
URL https://proceedings.mlr.press/v70/guo17a.html

[19] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, D. Kiela, Retrieval-augmented generation for knowledge-intensive NLP tasks, in: Advances in Neural Information Processing Systems, Vol. 33, 2020, pp. 9459–9474.
URL https://arxiv.org/abs/2005.11401

[20] R. Xu, G. Li, V. S. Sheng, GAVEL: Evidence-contract debate with mechanized scrutiny for provenance-grounded fact-checking, in: Findings of the Association for Computational Linguistics: ACL 2026, 2026. doi:10.18653/v1/2026.findings-acl.1789.
URL https://aclanthology.org/2026.findings-acl.1789/

[21] H. He, Y. Li, D. Wen, Y. Chen, R. Cheng, D. Chen, F. C. M. Lau, Debating truth: Debate-driven claim verification with multiple large language model agents, in: Proceedings of the ACM Web Conference 2026, 2026, arXiv:2507.19090. doi:10.1145/3774904.3792993.
URL https://arxiv.org/abs/2507.19090

[22] H. N. Truong, T. L. T. Nguyen, Q. V. H. Nguyen, T. P. T. Tran, M. A. Hoang, T. K. Dang, TERA-RAG: Temporal and epistemic reliability for multi-agent debate, in: 2026 23rd International Joint Conference on Computer Science and Software Engineering (JCSSE), 2026. doi:10.1109/JCSSE68839.2026.11596782.
URL https://doi.org/10.1109/JCSSE68839.2026.11596782

[23] H. Wang, X. Du, W. Yu, Q. Chen, K. Zhu, Z. Chu, L. Yan, Y. Guan, Learning to break: Knowledge-enhanced reasoning in multi-agent debate system, Neurocomputing 618 (2025) 129063. doi:10.1016/j.neucom.2024.129063.
URL https://www.sciencedirect.com/science/article/pii/S0925231224018344

[24] W. Yang, L. Some, M. Bain, B. Kang, A comprehensive survey on integrating large language models with knowledge-based methods, Knowledge-Based Systems 318 (2025) 113503. doi:10.1016/j.knosys.2025.113503.

[25] B. Yao, C. Shang, W. Du, J. He, R. Lian, Y. Zhang, H. Su, S. Swamy, Y. Qi, Peacemaker or troublemaker: How sycophancy shapes multi-agent debate, arXiv preprint arXiv:2509.23055 (2025).
URL https://arxiv.org/abs/2509.23055

[26] M. Sharma, M. Tong, T. Korbak, D. Duvenaud, A. Askell, S. R. Bowman, et al., Towards understanding sycophancy in language models, in: The Twelfth International Conference on Learning Representations, 2024, arXiv:2310.13548.
URL https://openreview.net/forum?id=tvhaxkMKAn

[27] Q. Wang, Z. Wang, Y. Su, H. Tong, Y. Song, Rethinking the bounds of LLM reasoning: Are multi-agent discussions the key?, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 6106–6131. doi:10.18653/v1/2024.acl-long.331.
URL https://aclanthology.org/2024.acl-long.331/

[28] E. Perez, S. Ringer, K. Lukosiute, K. Nguyen, E. Chen, S. Heiner, C. Pettit, C. Olsson, S. Kundu, S. Kadavath, et al., Discovering language model behaviors with model-written evaluations, in: Findings of the Association for Computational Linguistics: ACL 2023, Association for Computational Linguistics, 2023, pp. 13387–13434. doi:10.18653/v1/2023.findings-acl.847.

[29] M. Choi, K. Kim, S. Chae, S. Baek, An empirical study of group conformity in multi-agent systems, in: Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, 2025, pp. 5123–5139. doi:10.18653/v1/2025.findings-acl.265.

[30] A. Wynn, H. Satija, G. K. Hadfield, Talk isn't always cheap: Understanding failure modes in multi-agent debate, arXiv preprint arXiv:2509.05396 (2025).
URL https://arxiv.org/abs/2509.05396

[31] H. Ma, H. P. Zou, C. Li, E. Ma, Y. Su, P. S. Yu, Sycophancy suppression can impair rational updating: Anti-sycophancy should preserve the ability to update, in: Findings of the Association for Computational Linguistics: EMNLP 2026, Association for Computational Linguistics, 2026, pp. 25499–25515. doi:10.18653/2026.findings-emnlp.1208.

[32] T. Liang, Z. He, W. Jiao, X. Wang, Y. Wang, R. Wang, Y. Yang, S. Shi, Z. Tu, Encouraging divergent thinking in large language models through multi-agent debate, in: Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing, 2024, pp. 17889–17904.
URL https://aclanthology.org/2024.emnlp-main.992/

[33] C. Pham, B. Liu, Y. Yang, Z. Chen, T. Liu, J. Yuan, B. A. Plummer, Z. Wang, H. Yang, Let models speak ciphers: Multiagent debate through embeddings, in: The Twelfth International Conference on Learning Representations, 2024.
URL https://arxiv.org/abs/2310.06272

[34] Y. Zeng, W. Huang, L. Jiang, T. Liu, X. Jin, T. T. Chen, J. Li, X. Xu, S^2-MAD: Breaking the token barrier to enhance multi-agent debate efficiency, in: Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), 2025, pp. 9393–9408.
URL https://aclanthology.org/2025.naacl-long.475/

[35] A. P. Smit, N. Grinsztajn, P. Duckworth, T. D. Barrett, A. Pretorius, Should we be going MAD? A look at multi-agent debate strategies for LLMs, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235 of Proceedings of Machine Learning Research, PMLR, 2024, pp. 45883–45905.
URL https://proceedings.mlr.press/v235/smit24a.html

[36] Y. Li, Y. Du, J. Zhang, L. Hou, P. Grabowski, Y. Li, E. Ie, Improving multi-agent debate with sparse communication topology, in: Findings of the Association for Computational Linguistics: EMNLP 2024, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 7281–7294. doi:10.18653/v1/2024.findings-emnlp.427.
URL https://aclanthology.org/2024.findings-emnlp.427/

[37] Y. Liu, J. Cao, Z. Li, R. He, T. Tan, Breaking mental set to improve reasoning through diverse multi-agent debate, in: Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu (Eds.), International Conference on Learning Representations, 2025.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/3de667dab3b3d812583abc0a786139a0-Paper-Conference.pdf

[38] V. Subramaniam, Y. Du, J. B. Tenenbaum, A. Torralba, S. Li, I. Mordatch, Multiagent finetuning: Self improvement with diverse reasoning chains, in: The Thirteenth International Conference on Learning Representations, 2025.
URL https://arxiv.org/abs/2501.05707

[39] X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. H. Chi, S. Narang, A. Chowdhery, D. Zhou, Self-consistency improves chain of thought reasoning in language models, in: The Eleventh International Conference on Learning Representations, 2023.
URL https://arxiv.org/abs/2203.11171

[40] C.-M. Chan, W. Chen, Y. Su, J. Yu, W. Xue, S. Zhang, J. Fu, Z. Liu, ChatEval: Towards better LLM-based evaluators through multi-agent debate, in: The Twelfth International Conference on Learning Representations, 2024.
URL https://openreview.net/forum?id=FQepisCUWu

[41] G. M. Campedelli, N. Penzo, M. Stefan, R. Dessì, M. Guerini, B. Lepri, J. Staiano, I want to break free! Persuasion and anti-social behavior of LLMs in multi-agent settings with social hierarchy, Transactions on Machine Learning Research (2025).
URL https://openreview.net/forum?id=FR76oM8eGD

[42] K. Ueda, W. Hirota, T. Asakura, T. Omi, K. Takahashi, K. Arima, T. Ishigaki, Exploring the design of multi-agent LLM dialogues for research ideation, in: Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue, Association for Computational Linguistics, Avignon, France, 2025, pp. 322–337.
URL https://aclanthology.org/2025.sigdial-1.26/

[43] Y. Wang, K. Zhang, K. Tian, S. Zeng, X. Lv, N. Ding, B. Qi, B. Zhou, Scalability of LLM-based multi-agent systems for scientific code generation: A preliminary study, in: Proceedings of The 3rd Workshop on Mathematical Natural Language Processing (MathNLP 2025), Association for Computational Linguistics, Suzhou, China, 2025, pp. 50–61. doi:10.18653/v1/2025.mathnlp-main.4.
URL https://aclanthology.org/2025.mathnlp-main.4/

[44] S. Gómez Álvarez, A. Mozo Quesada, T. Navarro, S. Gálvez Rojas, F. López Valverde, Multi-agent debate system based on large language models: structured deliberation and validation in satellite communications, Journal of Intelligent Information Systems (Aug. 2026). doi:10.1007/s10844-026-01086-z.
URL https://doi.org/10.1007/s10844-026-01086-z

[45] D. Zhang, Z.-Z. Li, M.-L. Zhang, J. Zhang, Z. Liu, Y. Yao, H. Xu, J. Zheng, X. Chen, Y. Zhang, F. Yin, J. Dong, Z. Guo, L. Song, C.-L. Liu, From system 1 to system 2: A survey of reasoning large language models, IEEE Transactions on Pattern Analysis and Machine Intelligence 48 (3) (2026) 3335–3354. doi:10.1109/tpami.2025.3637037.
URL https://strathprints.strath.ac.uk/94892/7/Zhang-etal-IEEE-TPAMI-2025-From-system-1-to-system-2.pdf

[46] H. Zhou, H. Geng, X. Xue, L. Kang, Y. Qin, Z. Wang, Z. Yin, L. Bai, Reso: A reward-driven self-organizing llm-based multi-agent system for reasoning tasks, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2025, pp. 15979–15998. doi:10.18653/v1/2025.emnlp-main.808.
URL https://aclanthology.org/2025.emnlp-main.808

[47] C.-H. Chen, H.-H. Huang, H.-H. Chen, Self-augmented preference alignment for sycophancy reduction in LLMs, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Suzhou, China, 2025, pp. 12379–12391. doi:10.18653/v1/2025.emnlp-main.625.
URL https://aclanthology.org/2025.emnlp-main.625/

[48] L. Ibrahim, F. S. Hafner, L. Rocher, Training language models to be warm can reduce accuracy and increase sycophancy, Nature 652 (2026) 1159–1165. doi:10.1038/s41586-026-10410-0.

[49] Z. Gou, Z. Shao, Y. Gong, Y. Shen, Y. Yang, N. Duan, W. Chen, CRITIC: Large language models can self-correct with tool-interactive critiquing, in: The Twelfth International Conference on Learning Representations, 2024.
URL https://arxiv.org/abs/2305.11738

[50] P. Kushwaha, S. A. Madhavan, Agent-based contradiction detection and resolution for multi-source retrieval augmented generation systems, International Journal of Scientific Research in Engineering and Management (IJSREM) 10 (7) (Jul. 2026).
URL https://ijsrem.com/uploads/articles/1783285322570-ecd0e35bd42a-Research-paper-2.pdf

[51] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, W.-t. Yih, Dense passage retrieval for open-domain question answering, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Online, 2020, pp. 6769–6781. doi:10.18653/v1/2020.emnlp-main.550.
URL https://aclanthology.org/2020.emnlp-main.550/

[52] A. Estornell, Y. Liu, Multi-LLM debate: Framework, principals, and interventions, in: Advances in Neural Information Processing Systems, Vol. 37, 2024, pp. 28938–28964. doi:10.52202/079017-0911.
URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/32e07a110c6c6acf1afbf2bf82b614ad-Abstract-Conference.html

[53] Y. Cui, H. Fu, H. Zhang, L. Wang, C. Zuo, FREE-MAD: Consensus-free multi-agent debate, in: Findings of the Association for Computational Linguistics: ACL 2026, Association for Computational Linguistics, San Diego, California, United States, 2026, pp. 31977–31997. doi:10.18653/v1/2026.findings-acl.1600.
URL https://aclanthology.org/2026.findings-acl.1600/

[54] P. Pitre, G. Srivastava, L. Zhang, L. Wang, N. Ramakrishnan, X. Wang, A diagnostic study of multi-agent llms for real-world debates, in: Proceedings of the 43rd International Conference on Machine Learning, Vol. 306 of PMLR, Seoul, South Korea, 2026.
URL https://openreview.net/pdf?id=78Q4xkcJHc

[55] M. Cemri, M. Z. Pan, S. Yang, L. A. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, I. Stoica, Why do multi-agent LLM systems fail?, in: Advances in Neural Information Processing Systems, 2025, datasets and Benchmarks Track.
URL https://arxiv.org/abs/2503.13657

[56] E. Cau, V. Pansanella, D. Pedreschi, G. Rossetti, Selective agreement, not sycophancy: investigating opinion dynamics in LLM interactions, EPJ Data Science 14 (2025) 59. doi:10.1140/epjds/s13688-025-00579-1.

[57] X. Y. Lee, S. Akatsuka, L. Vidyaratne, A. Kumar, A. Farahat, C. Gupta, Reliable decision-making for multi-agent LLM systems, in: Workshop on Advancing LLM-Based Multi-Agent Collaboration at AAAI 2025, 2025.
URL https://multiagents.org/2025_artifacts/reliable_decision_making_for_multi_agent_llm_systems.pdf

[58] C. Zheng, H. Zhou, F. Meng, J. Zhou, M. Huang, Large language models are not robust multiple choice selectors, in: The Twelfth International Conference on Learning Representations (ICLR 2024), 2024.
URL https://openreview.net/forum?id=shr9PXz7T0

[59] S. Min, K. Krishna, X. Lyu, M. Lewis, W.-t. Yih, P. Koh, M. Iyyer, L. Zettlemoyer, H. Hajishirzi, FActScore: Fine-grained atomic evaluation of factual precision in long form text generation, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2023, pp. 12076–12100. doi:10.18653/v1/2023.emnlp-main.741.
URL https://aclanthology.org/2023.emnlp-main.741/

[60] J. Pineau, P. Vincent-Lamarre, K. Sinha, V. Larivière, A. Beygelzimer, F. d'Alché Buc, E. Fox, H. Larochelle, Improving reproducibility in machine learning research (a report from the NeurIPS 2019 reproducibility program), Journal of Machine Learning Research 22 (164) (2021) 1–20.
URL https://jmlr.org/papers/v22/20-303.html

[61] L. Phan, A. Gatti, Z. Han, N. Li, J. Hu, et al., A benchmark of expert-level academic questions to assess AI capabilities, Nature 649 (2026) 1139–1146, arXiv:2501.14249; Humanity's Last Exam. doi:10.1038/s41586-025-09962-4.

[62] M. Sarrouti, A. Ben Abacha, Y. Mrabet, D. Demner-Fushman, Evidence-based fact-checking of health-related claims, in: Findings of the Association for Computational Linguistics: EMNLP 2021, Association for Computational Linguistics, Punta Cana, Dominican Republic, 2021, pp. 3499–3512. doi:10.18653/v1/2021.findings-emnlp.297.
URL https://aclanthology.org/2021.findings-emnlp.297/

[63] I. Petrov, J. Dekoninck, M. Vechev, BrokenMath: A benchmark for sycophancy in theorem proving with LLMs, arXiv preprint arXiv:2510.04721 (2025).
URL https://arxiv.org/abs/2510.04721

[64] J. Dekoninck, N. Jovanovic, T. Gehrunger, K. Rögnvaldsson, I. Petrov, C. Sun, M. T. Vechev, Beyond benchmarks: MathArena as an evaluation platform for mathematics with LLMs, arXiv preprint arXiv:2605.00674 (2026).
URL https://arxiv.org/abs/2605.00674

[65] J. Cohen, A coefficient of agreement for nominal scales, Educational and Psychological Measurement 20 (1) (1960) 37–46. doi:10.1177/001316446002000104.

[66] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, D. T. Mellor, The preregistration revolution, Proceedings of the National Academy of Sciences 115 (11) (2018) 2600–2606. doi:10.1073/pnas.1708274114.

[67] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.
URL https://www.jstor.org/stable/4615733

[68] Q. McNemar, Note on the sampling error of the difference between correlated proportions or percentages, Psychometrika 12 (2) (1947) 153–157. doi:10.1007/BF02295996.

[69] B. Efron, R. J. Tibshirani, An Introduction to the Bootstrap, Chapman & Hall/CRC, New York, 1993. doi:10.1201/9780429246593.

[70] P. Koehn, Statistical significance tests for machine translation evaluation, in: Proceedings of the 2004 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Barcelona, Spain, 2004, pp. 388–395.
URL https://aclanthology.org/W04-3250/

[71] D. Card, P. Henderson, U. Khandelwal, R. Jia, K. Mahowald, D. Jurafsky, With little power comes great responsibility, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Online, 2020, pp. 9263–9274. doi:10.18653/v1/2020.emnlp-main.745.